ST 3007
Operational Research

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Course information

- **Lecture Hours**: Monday 3.00 pm – 4.00 pm (SCH)

  Tuesday 1.00 pm – 3.00 pm (SLT-A)

- **Evaluation Criteria**:
  - End-Semester Examination (70%)
  - Continuous Assessments (30%): Two In-class Assignments

- **Main Reference:**

  Taha, H. A. (1997). Operations research: an introduction (6th ed).Prentice Hall, Upper Saddle River, N.J

Learning Outcomes

After the course you will be able to:

- Describe the fundamental concepts of real-world applications in operational research.
- Model decision making problems.
- Obtain solution/s for the formulated model/s using appropriate techniques.
- Use of software packages to solve operational research problems.

Notional Hours - 150 hours

Activity	Hours
Lectures	45 hours
Tutorials	15 hours
Lab work	15 hours
Group work	15 hours
Other (Self Studies)	60 hours
Total Notional Hours	150 hours

Integer Programming

Chapter 1

Chapter Summary

- Motivation
- What is Integer Programming?
- Gomory's Cutting Plane Algorithm

Motivation

- In Linear Programming (LP) problems, decision variables are restricted to be **non-negative** (greater than or equal to zero).
- Further, decision variables are **continuous** (property of continuity). Hence, fractional values for decision variables are possible in solutions of LP problems.
- There are LP problems that the assumption of continuity is valid. E.g.: production of different fertilizers (in kg or tons) in a LP model on production mix, usage of different amounts of food items (in grams) in a LP model on balanced diet, etc.
- However, there are many LP problems where decision variables are people, vehicles, machines, etc. which cannot be assigned in fractions.
- This means that in these LP problems, those decision variables should take integer values only.

What is Integer Programming?

- Since there are many instances where decision variables can only consist with integer values, methods to incorporate this integer constraint is required.
- The mathematical model for Integer Programming (IP) or more precisely Integer Linear Programming (ILP) is LP models with additional constraint/restriction; "*variables must have integer values*".
- If only some of the variables of the problem should be integers, then they are called "**Mixed IP Problems (MIP)".**
- If all the variables of the problem should take integer values, then we have "**Pure IP Problems (PIP**)".

Gomory's Cutting Plane Algorithm

- One method to solve *IP problems* is **Gomory's cutting plane algorithm**, however, it can be **only** used to solve *pure IP problems*.
- The general procedure for cutting plane method is as follows:

Step 1: Find the solution to the original LP problem (using the simplex method) ignoring the integer constraint. This is called the **relaxation** of the IP problem.

Step 2: If the solution is integer stop. Otherwise, construct a "**cut**" derived from the row that has a non-integer variable with **the largest fraction value** and add it to the current final tableau. If there is a tie in the largest fraction, select any row **arbitrarily**.

Step 3: Solve the augmented LP problem and return to step 2.

Example: Cutting Plane Algorithm

- Consider the following ILP problem.

$$\textbf{Maximize } Z = 7x_1 + 10x_2$$

Subject to:

$$-x_1 + 3x_2 \leq 6$$

$$7x_1 + x_2 \leq 35$$

$$x_1, x_2 \geq 0 \text{ and integers}$$

- Above LP problem is a Pure IP problem as all the variables should be integers.
- Therefore, we can use Gomory's cutting plane algorithm to solve the above problem.

Example: Cutting Plane Algorithm

- The first step is to derive the **optimal** solution by ignoring integer constrain using the simplex method.

Exercise: Derive the optimal tableau for the LP problem using the simplex method.

Example: Cutting Plane Algorithm Solution

- You should obtain the Optimal tableau as below.

Basic	X_1	X_2	S_1	S_2	Solution
z	0	0	-63/22	-31/22	**-133/2**
X_2	0	1	7/22	1/22	$\mathbf{3\frac{1}{2}}$
X_1	1	0	-1/22	3/22	$\mathbf{4\frac{1}{2}}$

Example: Cutting Plane Algorithm Solution

- The basic variable equations from the optimum tableau can be written explicitly as below;

$$X_2 + 7/22S_1 + 1/22S_2 = 3\frac{1}{2}$$

$$X_1 - 1/22S_1 + 3/22S_2 = 4\frac{1}{2}$$

- Pick the basic variable equation with **the largest fractional value** to deduce the fractional cut.
- **Note:** It doesn't matter which equation is chosen to the final solution, however *the number of iterations is less when choose the equation with the largest fraction*.

Example: Cutting Plane Algorithm Solution

- In this case, since both equation's fractional values are same, we will just choose the equation of X_2.

$$X_2 + 7/22S_1 + 1/22S_2 = 3\frac{1}{2}$$

- Rewrite the equation so that the all the non integer coefficients are written as an integer value and strictly positive fractional component.

$$X_2 + \left(0 + \frac{7}{22}\right)S_1 + \left(0 + \frac{1}{22}\right)S_2 = 3 + \frac{1}{2}$$

- Keep integer values and variables with integer coefficients in the left-hand side and move the fractions and variables with fractional values to right hand side.

$$X_2 + 0S_1 + 0S_2 - 3 = -\frac{7}{22}S_1 - \frac{1}{22}S_2 + \frac{1}{2} \quad (1)$$

Example: Cutting Plane Algorithm Solution

- Since the S_1 and S_2 are non-negative, right-hand side of the equation (1) should be less than ½;

$$-\frac{7}{22}S_1 - \frac{1}{22}S_2 + \frac{1}{2} \leq \frac{1}{2}$$

- Note that for derivation of the cut, we consider all the variables to be integers (including S_1 and S_2), hence left-hand side of the equation (1) is an integer so the right-hand side will also be an integer.

- Therefore, if an integer is at most ½, then it is actually at most 0. Hence the right-hand side of the equation (1) should be less than 0;

$$-\frac{7}{22}S_1 - \frac{1}{22}S_2 + \frac{1}{2} \leq 0$$

$$-\frac{7}{22}S_1 - \frac{1}{22}S_2 \leq -\frac{1}{2} \qquad (2)$$

Example: Cutting Plane Algorithm Solution

- The equation (2) is added as a new constraint (cut) to the LP optimum tableau.
- Resulting tableau is **optimal but infeasible**. Therefore, we can implement dual simplex method. We look for most negative value in solution column and get the pivot row, then implement minimum test on the negative values of pivot row.

Basic	X_1	X_2	S_1	S_2	S_{g1}	Solution
z	0	0	-63/22	-31/22	0	**-133/2**
X_2	0	1	7/22	1/22	0	$3\frac{1}{2}$
X_1	1	0	-1/22	3/22	0	$4\frac{1}{2}$
S_{g1}	0	0	$\boxed{-7/22}$	-1/22	1	$-\frac{1}{2}$

smallest absolute value $\underbrace{\left|\frac{-63/22}{-7/22}\right|}$ $=9$ $\left|\frac{-31/22}{-1/22}\right|=31$

Example: Cutting Plane Algorithm Solution

- Applying the dual simplex method to recover feasibility yield;

Basic	X_1	X_2	S_1	S_2	S_{g1}	Solution
z	0	0	0	-1	-9	**-62**
X_2	0	1	0	0	1	**3**
X_1	1	0	0	1/7	-1/7	$\mathbf{4\frac{4}{7}}$
S_1	0	0	1	1/7	-22/7	$\mathbf{1\frac{4}{7}}$

- This solution is still non integer in X_1 and S_1. To derive the cut, we will use X_1 equation arbitrarily. Cut is given as below.

$$-\frac{1}{7}S_2 - \frac{6}{7}S_{g1} \leq -\frac{4}{7}$$

Example: Cutting Plane Algorithm Solution

- After adding the new cut resulting tableau is given below. We apply dual simplex method here also.

Basic	X_1	X_2	S_1	S_2	S_{g1}	S_{g2}	Solution
z	0	0	0	-1	-9	0	**-62**
X_2	0	1	0	0	1	0	**3**
X_1	1	0	0	1/7	-1/7	0	$\mathbf{4\frac{4}{7}}$
S_{g1}	0	0	1	1/7	-22/7	0	$\mathbf{1\frac{4}{7}}$
S_{g2}	0	0	0	$\boxed{-1/7}$	-6/7	1	$\mathbf{-\frac{4}{7}}$

$$\left|\frac{-1}{-1/7}\right| = 7 \qquad \left|\frac{-9}{-6/7}\right| = 10.5$$

Example: Cutting Plane Algorithm Solution

- The dual simplex method yields the following tableau which is optimal, feasible and integer.

Basic	X_1	X_2	S_1	S_2	S_{g1}	S_{g2}	Solution
z	0	0	0	0	-3	-7	**-58**
X_2	0	1	0	0	1	0	**3**
X_1	1	0	0	0	-1	1	**4**
S_{g1}	0	0	1	0	-4	1	**1**
S_{g2}	0	0	0	1	6	-7	**4**

- Final solutions; $\boldsymbol{X_1 = 4, X_2 = 3,\ Max\ Z = 58}$

Cutting Plane Algorithm - Remarks

- It is important to note that the fractional cut assumes that all the variables including the slack and surplus variables are integer.
- This means that the cut deals with pure integer problems only.

Thank You!

Next Chapter - Mixed Integer Programming using Branching and Bounding method